

Realsock - XAI Supply Chain-to-Housing Market Radar - Research Impact Map

Product case: Hormuz alert - from live logistics signal to a traceable Australian project decision

What is the problem my research addresses:

write in lay language. watch for jargon

A builder receives a shipping alert but cannot immediately tell whether it affects a named project, package or delivery gate. The information is fragmented across news, supplier emails, freight quotes and spreadsheets.

Realsock Radar closes this gap: it monitors a supply-chain event, links it to project exposure, simulates plausible cost/time outcomes, and shows the next action with its evidence trail.

Evidence to illustrate the problem:

facts and figures to make the problem relatable

Live demo - six-week Hormuz scenario:

event confidence **0.82**; route delay **+7-14 sea days**; freight **+18-45%**; cargo cover **+0.6-1.4pp**.

For an illustrative **A\$80m** apartment build with an **A\$8.6m** fittings package on the critical path, the Radar tests a **+1.8-4.6%** build-cost range (**A\$1.44-3.68m**) and **+2-8 weeks** programme range. **Housing Purchase Capacity Index: 96-99 / 100** in this case. These are scenario outputs, not confirmed forecasts.

Inputs: IEA Hormuz factsheet (Feb 2026); ABS PPI (Dec 2025); user-defined project assumptions.

Why does someone care about the problem?

motivations to connect

A decision made late can turn a manageable freight issue into a variation, idle labour or a missed handover. The Radar gives a project team a **48-hour** response window: verify the route, obtain two landed-cost quotes, and check stock or substitutes before releasing the **A\$8.6m** fittings package.

It replaces a vague alert with a decision trigger: **if ETA slips beyond 14 days at week 6, activate the adverse plan and re-test contingency.**

Who might be that someone?

specificity is good

First user: Australian residential developers and builders with imported materials on a critical path.

Direct users: procurement manager, project manager, commercial manager and finance lead. **Contributors:** supplier, freight forwarder, insurer and quantity surveyor.

Users receive: a live alert, a project-specific exposure check, scenario range, owner/due-date task list and exportable decision record.

The part of the problem I'm focused on:

this is your research topic

A practical, explainable question: **when a supply-chain event is detected, which active housing-project commitments are exposed, by how much, and what should the team do next?**

The initial Radar focuses on direct pathways that can be checked: event -> route -> freight/cover -> imported package -> cost, cash flow and programme. Housing-price effects remain conditional, longer-horizon outputs.

How I propose to solve the problem:

this is your research approach

Realsock Radar workflow:

- Detect:** monitor and classify an event; retain source and confidence.
- Connect:** traverse the context graph to routes, suppliers, materials and project milestones.
- Simulate:** compare normal, watch and adverse ranges across cost, delivery and cash flow.
- Explain:** reveal every link, assumption, conflict and supporting source.
- Act:** assign an owner, due date and trigger for quote, substitute, contingency or reforecast.

Housing Purchase Capacity Index (base = 100) translates the selected scenario into three visible channels: new-dwelling cost, mortgage repayment pressure and settlement timing. The implementation uses Semantica-style context graphs and deterministic paths, not a black-box score.

All these people care about my research (before, during, after):

your end/next users, partners, beneficiaries, media comms, your HoS

Before: developers provide anonymised package, supplier, budget and milestone data; logistics partners define route and quote checks.

During: the Radar combines source records with the project context graph. Users can challenge a connection, change a scenario assumption, or compare a trade-diversion or RBA +25bp case.

After: each decision is retained as a replayable, exportable record for the project team, lender, insurer or governance review.

How they'll know about my research:

outputs, communication, and dissemination approach

Product outputs in the live Radar:

- event confidence and source record; **13 nodes / 15 causal pathways / 5 decision checks** in the Hormuz case;
- budget-at-risk breakdown: freight **26%**, cargo cover **13%**, materials/delay **61%**;
- six-week scenario fan: normal **0-0.8%**, watch **0.9-1.7%**, adverse **1.8-4.6%**;
- purchase-capacity index: Hormuz **96-99**, trade diversion **97-99**, RBA +25bp **92-95** (base = 100);
- critical-path clock: D+2 route confirmation, D+5 stock/substitute, D+7 quote lock, D+10 PM decision, week-6 reforecast.

The QR opens a phone-ready interactive demo; the decision record can be shared with project stakeholders.

The impact it'll generate:

because of all of this, x will be better

Why this is different: news platforms flag an event; property dashboards report lagging outcomes; black-box tools output a risk score. **Realsock Radar connects the three:** live signal -> named project exposure -> transparent simulation -> assigned action. Its purchase-capacity index makes the household-facing implication legible without pretending to forecast house prices: **100 = baseline; lower values mean a scenario makes a new home less affordable through cost, repayments or settlement timing.**

Business outcome to test: faster exposure verification, fewer unsupported alerts, earlier quote/substitution decisions, clearer lender/supplier communication and documented contingency choices. Funding delivers a low-cost, mobile-ready, explainable planning product for Australian housing projects.



OPEN THE LIVE DEMO

Scan to explore the interactive Hormuz case study

anhdatle.github.io/realsock-xai-simulator

Mobile-ready XAI simulation and decision trace